

EDUCATION

University of California-Irvine , Irvine, CA	Class of 2027
- B.S. Mechanical Engineering, B.S. Aerospace Engineering - Materials Science and Engineering, Minor - Tau Beta Pi, Engineering Honor Society Member; Sigma Gamma Tau, National Aerospace Engineering Honor Society Member - Certified SolidWorks Associate (Mechanical Design)	GPA 3.75

EXPERIENCE | WORK, VOLUNTEER, & LEADERSHIP

Toyota Racing Development USA Mechanical Engineering Intern, <i>Costa Mesa, CA</i>	06.2026-09.2026
Conducted engine component failure analysis and materials testing, included optical/confocal/SEM microscopy, Rockwell hardness testing, metallographic sample preparation, and re-commission, calibration, and standardization of an RDE-OES oil analysis spectrometer.	
Advanced Casting Research Center Valdevit Research Group, <i>Irvine, CA</i>	07.2026 - present
Research on Cold Spray Additive Manufacturing, investigating material deposition behavior and process parameters to support the development of advanced manufacturing methods.	
MD Anderson Cancer Center Physics and Engineering Research Intern, <i>Houston, TX</i>	06.2025 - 08.2025
Developed an automated machine learning model to classify imagery data using radiomic features from CT scans. Implemented feature selection (LASSO, Sequential Backward Selection) & trained classifiers (KNN, Neural Networks) to improve assessment.	
MotoGP Track Marshal, Circuit of the Americas, <i>Austin, TX</i>	03.2025 - present
Served as a track marshal for the 2025 MotoGP through the Sports Car Club of America (SCCA) and MotorsportReg, supporting real-time incident response. Gained experience in real-time system operations, verbal communication, and critical thinking.	
UC Irvine Parking and Transportation Services Anteater Express Bus Driver, <i>Irvine, CA</i>	08.2024 - present
Class B licensed Anteater Express bus driver. Responsible for the inspection and safe operation of shuttle service buses at UC Irvine. Responsible for the safety and well-being of 250+ passengers per shift.	
Engineering Student Council Faculty Engagement Committee, <i>Irvine, CA</i>	10.2023 - present
Promote student-faculty interaction for the educational, social, and professional advancement of the engineering community at UCI. Create networking and research opportunities with other organizations, businesses, and faculty.	
Galileo Camps Lead Instructor, <i>Irvine, CA</i>	06.2022 - 08.2024
Led STEM sessions for children, designing hands-on projects and lesson plans to enhance learning. Fostered an inclusive environment, inspiring creativity and curiosity in campers.	

PROJECTS

Solar Car Suspension & Steering Lead, <i>UCI Solar Racing, Zotsun Solar Car Senior Design Project</i>	08.2025 - present
Contributing to the design and development of a solar car for the Formula Solar Grand Prix. Working on suspension and steering systems, creating CAD models, applying GD&T to ensure precise manufacturing, analyzing mechanical components, and assisting in the fabrication, machine operation, and assembly of steering and suspension systems.	
Curtiss JN-4 Biplane Fuselage Engineer, <i>Flying Leatherneck Aviation Museum</i>	03.2025 - present
Collaborating to build a scale replica of the Curtiss JN-4 "Jenny" biplane, interpreting 2D engineering drawings, analyzing structural components, and assisting in the hands-on fabrication, machine operation, and assembly of the fuselage.	
Unmanned Aerial Vehicle Fuselage Engineer, <i>Fixed-Wing Innovation Project Club</i>	01.2025 - present
Part of a multidisciplinary team, designing, 3D printing, and flight testing a fixed wing aircraft for an aerospace competition focused on additive manufacturing, flight performance optimization, and structural integrity.	
Transit Bus Design: CAD Assembly & FEA Analysis Class Project, <i>Computer-Aided Design</i>	09.2024 - 12.2024
Designed CAD model of a transit bus in SolidWorks with motion and performed FEA analysis to find maximum loading.	
Autonomous Robot & RC Rover Class Projects, <i>Mechanical Systems Laboratory</i>	09.2023 - 03.2024
- Utilized Arduino programming in C, 3D printing, and laser cutting to construct an autonomous rover. Applied mechanical system design and propulsion testing with electro-mechanical systems (magnetometer, solenoid, limit switch, piston, servos, motors). - Built and designed a rover and its steering system using a servo motor to control the steering system and a receiver/transmitter to control the rover.	
Website Independent Project, <i>anmassoumi.github.io</i>	03.2024 - present
Using HTML and CSS, designed and coded website on VS Code to showcase my projects.	

SUMMARY

- Software:** CAD, FEA Simulation, MATLAB, Arduino, CSS, HTML, Python, Data Processing, MS Office
- Hardware:** 3D Printing, Laser Cutting, Fabrication Machinery Shop Equipment
- Additional:** Machine Learning Coding, Communication Skills, Analytical & Problem Solving Capabilities, Passion for motorsport engineering and advancing automotive performance through engine design, development, and production utilizing dynos